

Regulation of Gene Expression - Factoid Sheet

Week 4 - Thursday - Core Lecture 11 - current faculty deck + official Week 4 objectives

Master chain: cis sequence + trans factor → chromatin accessibility → transcription → RNA processing/export → translation → protein control.

Gene-Expression Language

Cis = local DNA/RNA sequence: promoter, enhancer, silencer/operator, response element.

Trans = diffusible protein/ncRNA: TF, activator, repressor, coactivator, miRNA, lncRNA.

Lac Operon

Signal	Effect
Lactose present	Allolactose binds repressor → operator freed
Glucose low	cAMP rises → CRP/CAP activation → stronger RNAP recruitment
Lactose + / glucose low	Maximal transcription

Eukaryotic Levels

Genome/chromatin • transcription • RNA processing/export • translation • post-translational control.

Epigenetics

Change	High-yield result
HAT / histone acetylation	Open chromatin; expression up
HDAC	Compaction; expression down
Histone methylation	Context dependent
DNA CpG methylation	Usually compaction / silencing

Imprinting

PWS = active paternal 15q11-q13 contribution lost. Paternal deletion or maternal UPD.

Angelman = active maternal contribution lost. Maternal deletion or paternal UPD.

Gene Regulation - Fast Recall

Transcription Factors

Modular: DNA-binding domain + activation/regulatory domain + NLS; nuclear receptors also have ligand-binding and dimerization domains.

Four DBD motifs	Remember
Helix-turn-helix	Helical recognition
Helix-loop-helix	DNA binding + dimerization
Leucine zipper	Dimerization interface
Zinc finger	Steroid/nuclear receptor motif

Estrogen Receptor

Type I nuclear receptor: estrogen binds intracellular ER → dimerization/nuclear entry → ERE binding → coactivator-mediated transcription. Tamoxifen blocks the productive coactivator conformation.

RNA Interference

	miRNA	siRNA
Pairing	Usually partial	Usually full
Targets	Many possible	More specific
Main outcome	Translation repression / degradation	mRNA degradation
Shared	RISC	

X-Inactivation

XIST coats one X → hypermethylation + histone deacetylation → Barr body. Random selection creates mosaicism. Some genes, especially pseudoautosomal genes, escape.

Must-Not-Miss

Cis = sequence; trans = factor. Lactose removes the brake, low glucose adds the accelerator. HAT opens. DNA methylation usually silences. PWS paternal / AS maternal. HTH-HLH-zipper-zinc finger. miRNA/siRNA use RISC. XIST silences X.

Assigned ScholarRx

Epigenetics: <https://exchange.scholarrx.com/brick/epigenetics>
Genomic Imprinting Disorders: <https://exchange.scholarrx.com/brick/genomic-imprinting-disorders>